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Tutorial 6

Question:

A fashion company wants to automatically generate new clothing designs using AI. They have a large dataset of clothing images such as dresses, shirts, and shoes. The company plans to use Generative Models like GANs and VAEs to learn patterns from existing designs and create new realistic clothing images.

1. What type of machine learning model should the company use to generate new clothing designs? Explain the concept and working of GAN for this application. 2. If the company wants to learn compressed representations of clothing features and generate new designs, how can Variational Autoencoders (VAEs) help? Explain their architecture and working.

3. Which model would produce more realistic clothing designs for the company, GAN or VAE? Compare both models and mention their applications.

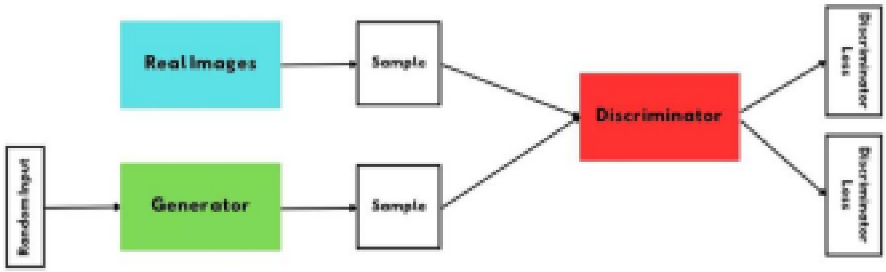
Answer:

1. Generative Models and Working of GAN in Clothing Design Generation. Company should use **Generative Models**, to learn patterns of existing data & generate new data.

One such model is **GAN (Generative Adversarial Network)**.

Architecture of GAN

- **Generator** – generates new clothing images.
- **Discriminator** – checks whether the image is real or generated.



Working:

1. Generator creates fake clothing images.
2. Discriminator compares fake images with real images.
3. Discriminator detects fake images.
4. Generator improves its output.
5. Process repeats until generated images look realistic.

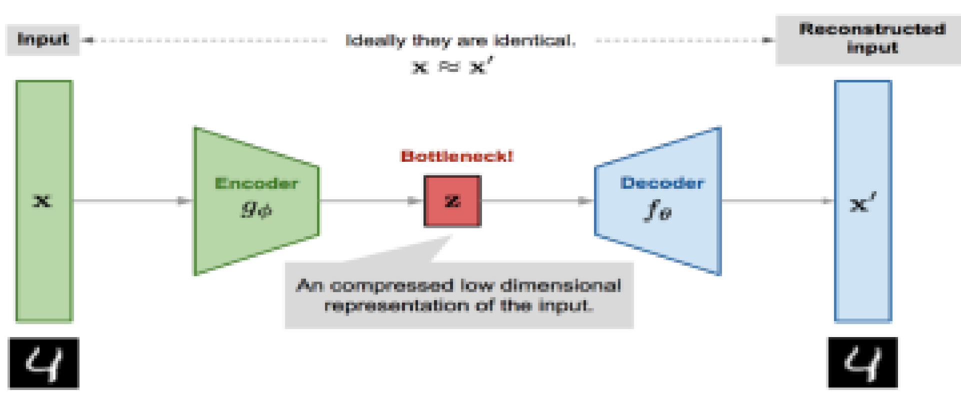
2. Architecture and Working of Variational Autoencoders (VAEs) for Feature Learning.

A **Variational Autoencoder (VAE)** can learn hidden features of clothing designs and generate new images.

Architecture:

• **Encoder** – compresses clothing images into latent variables.

• **Latent Space** – stores important features like color, pattern, and shape. •
Decoder – generates new clothing images from latent space.



Working:

1. Clothing image is given as input.

2. Encoder converts it into latent representation.
3. Random samples are taken from latent space.
4. Decoder generates new clothing designs.

Latent Space:
A compressed representation where important features of clothing designs are stored.

3. Comparison of GAN and VAE for Realistic Image Generation and Their Applications
GANs usually produce more realistic images, while VAEs provide stable learning and structured representations.

Feature	GAN	VAE
Architecture	Generator + Discriminator	Encoder + Decoder
Output Quality	Highly Realistic	Slightly Blurry
Training	Difficult	Easier

Applications:
- Fashion design generation
- Face generation
- Medical image creation
- Image enhancement
- Data augmentation